

Aim: Exceptional Handling.**1. Write a java program to illustrate basic of exceptional handling.****CODE:**

```
import java.util.Scanner;
public class TryCatch {
    public static void main(String args[]){
        double r;
        int x,y;
        try{
            Scanner s = new Scanner(System.in);
            System.out.print("Enter x:");
            x = s.nextInt();
            System.out.print("Enter y:");
            y = s.nextInt();
            r = x/y;
        }
        catch(ArithmeticException ae){
            System.out.println("Not divisible by 0");
        }
        catch(Exception e){
            System.out.println("Some Arithmetic error has occurred!");
        }
        finally{
            System.out.println("Finally is always executed ;");
        }
    }
}
```

OUTPUT:

```
Enter x:200
Enter y:100
Finally is always executed ;)
```

OUTPUT of ArithmeticException:

```
Enter x:100
Enter y:0
Not divisible by 0
Finally is always executed ;)
```

2. Write a java program to handle division by zero exception using random class.**CODE:**

```
import java.util.Random;
public class HandlingRandom {
    public static void main(String args[]){
        try{
            Random r = new Random();
            int n1 = r.nextInt(10);
            int n2 = r.nextInt(5);
            System.out.println("1st random number:" + n1);
            System.out.println("2nd random number:" + n2);
            int res = n1/n2;
        }
    }
}
```

```
        System.out.println("Division of random number is " + res);
    }
    catch(ArithmeticException ae){
        System.out.println("Can't dividde by Zero!");
    }
}
}
```

OUTPUT:

1st random number:4
2nd random number:2
Division of random number is 2

OUTPUT of ArithmeticException:

1st random number:0
2nd random number:0
Can't dividde by Zero!

3. Write a java program to illustrate how we can write multiple try-catch finally block in a single program.

CODE:

```
import java.util.Scanner;
public class MultipleTryCatch{
    public static void main(String args[]) {
        Scanner s = new Scanner(System.in);
        int[] num = null;
        int n = 0;
        try {
            System.out.print("Enter number you want:");
            n = s.nextInt();
            num = new int[n];
            System.out.print("Enter the number:");
            for (int i = 0; i < n; i++) {
                num[i] = s.nextInt();
            }
            System.out.print("Enter the number for the denominator:");
            int ch = s.nextInt();
            for (int i = 0; i < n; i++) {
                num[i] = num[i] / ch;
            }
            System.out.print("Array of elements after division:");
            for (int i = 0; i < n; i++) {
                System.out.print(num[i] + " ");
            }
        }
        catch (ArithmeticException ae) {
            System.out.println("\nCan't divide by zero!");
        }
        finally {
            System.out.println("\nDivision operation completed.");
        }
    }
}
```

```
try {
    System.out.print("Enter the index number to access the element in the array:");
    int in = s.nextInt();
    System.out.println("Number present at index " + in + ":" + num[in]);
}
catch (ArrayIndexOutOfBoundsException ai) {
    System.out.println("\nGiven index is not valid!!");
}
finally {
    System.out.println("\nArray access operation completed.");
    s.close();
}
}
```

OUTPUT:**OUTPUT of ArithmeticException:**

Enter number you want:5
Enter the number:10 20 30 40 50
Enter the number for the denominator:0
Can't divide by zero!

Division operation completed.

Enter the index number to access the element in the array: 2

Number present at index 2:30

Array access operation completed.

OUTPUT OF ArrayIndexOutOfBoundsException:

Enter number you want:5
Enter the number:10 20 30 40 50
Enter the number for the denominator:5
Array of elements after division:2 4 6 8 10
Division operation completed.

Enter the index number to access the element in the array:10

Given index is not valid!!

Array access operation completed.

4. Write a java program to demonstrate one try block can have multiple catch blocks.**CODE:**

```
import java.util.Scanner;
public class MultipleCatch {
    public static void main(String args[]){
        Scanner s = new Scanner(System.in);
        try{
            System.out.print("Enter number you want:");
            int n = s.nextInt();
            int num[] = new int[n];
            System.out.print("Enter the number:");
            for(int i=0;i<n;i++){
                num[i] = s.nextInt();
            }
        }
    }
}
```

```
}
System.out.print("Enter the number for the denominator:");
int ch = s.nextInt();
for(int i=0;i<n;i++){
    num[i] = num[i]/ch;
}
System.out.print("Array of element after division:");
for(int i=0;i<n;i++){
    System.out.print(num[i]+" ");
}
System.out.print("\nEnter the index number to access the element in the array:");
int in = s.nextInt();
System.out.println("Number present at index " + in + ":" + num[in]);
}
catch(ArithmeticException ae){
    System.out.println("Can't divide by zero!");
}
catch(ArrayIndexOutOfBoundsException ai){
    System.out.println("Given index is not valid!!");
}
catch(NegativeArraySizeException na){
    System.out.println("Warning!, Negative size is not allowed!!");
}
}
}
```

OUTPUT:**OUTPUT OF NegativeArraySizeException:**

Enter number you want:-5

Warning!, Negative size is not allowed!!

OUTPUT OF ArrayIndexOutOfBoundsException:

Enter number you want:5

Enter the number:10 20 30 40 50

Enter the number for the denominator:5

Array of element after division:2.0 4.0 6.0 8.0 10.0

Enter the index number to access the element in the array:10

Given index is not valid!!

OUTPUT OF ArithmeticException:

Enter number you want:5

Enter the number:10 20 30 40 50

Enter the number for the denominator:0

Can't divide by zero!

5. Write a java program to demonstrate the use of Throwable class, by illustrating that the object of class that extends Throwable can be thrown and caught. (Hint: Note that Exception is subclass of Throwable and thus create a user defined class MyException that will extend Throwable class)

CODE:

```
class MyException extends Exception{
    MyException(String message){
        super(message);
    }
}
public class TestException{
    public static void main(String[] args) {
        int a=60;
        int b=70000;
        try{
            float r = (float)a/(float)b;
            if(r<0.1){
                throw new MyException("The resultant number is too small");
            }
        }
        catch(MyException e){
            System.out.println("Exception is caught!");
            System.out.println(e.getMessage());
        }
        finally{
            System.out.println("I can't be interrupted :)");
        }
    }
}
```

OUTPUT:

```
Exception is caught!
The resultant number is too small
I can't be interrupted :)
```